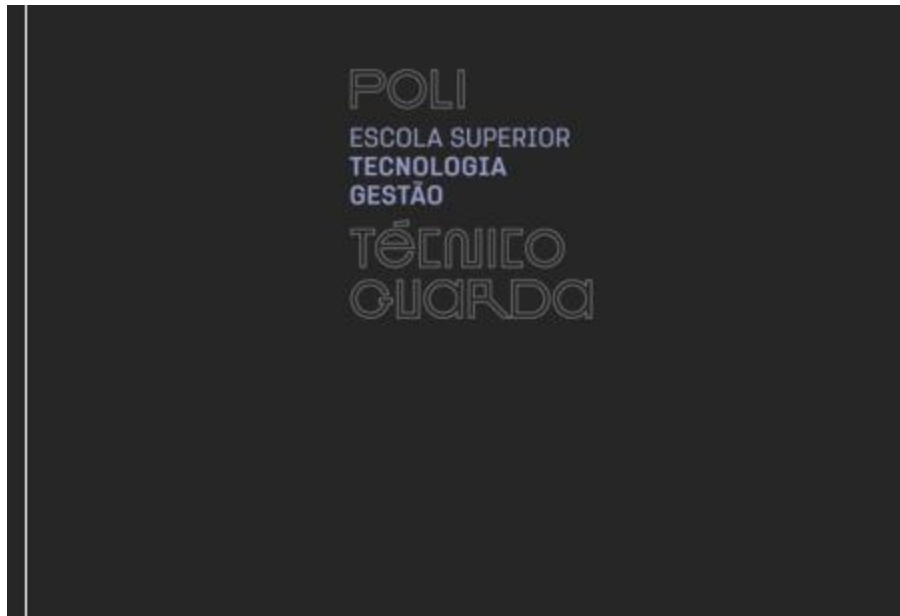
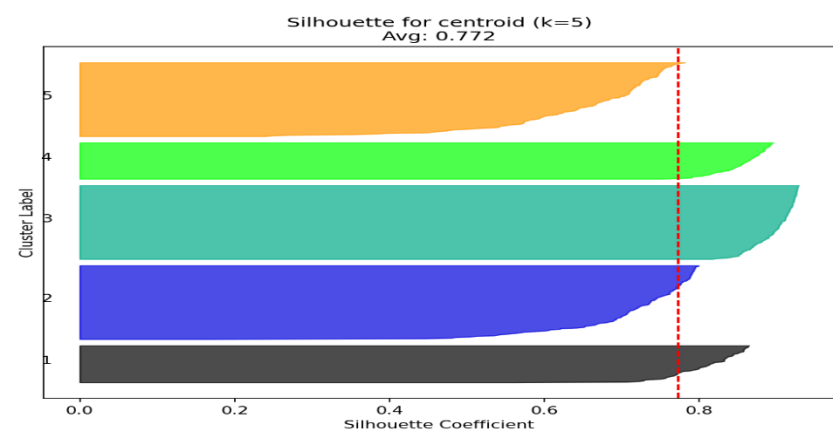
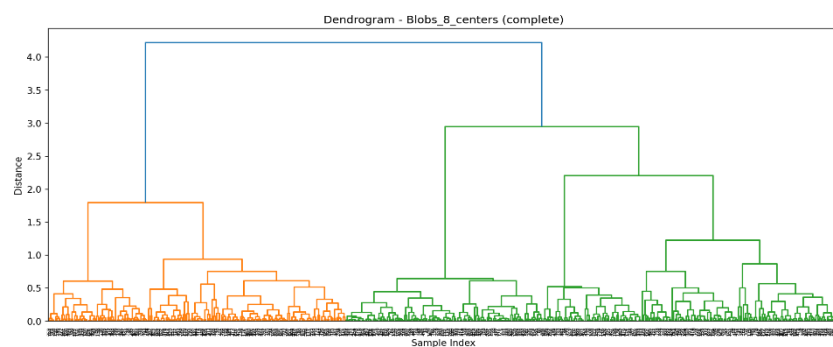
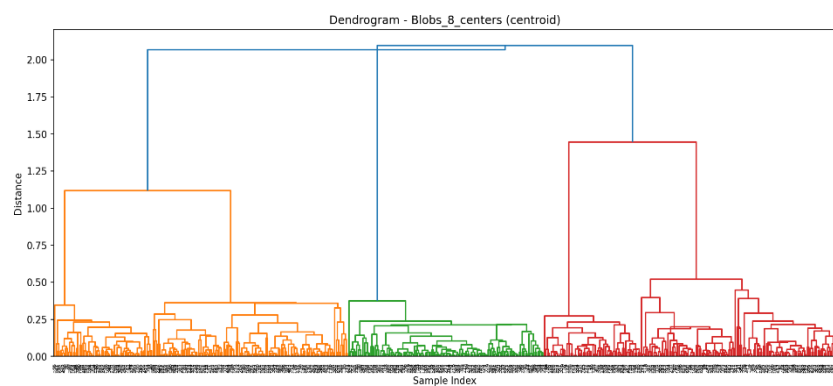
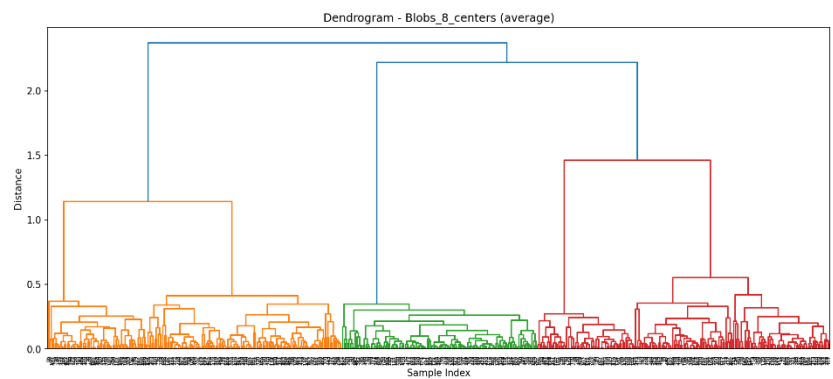


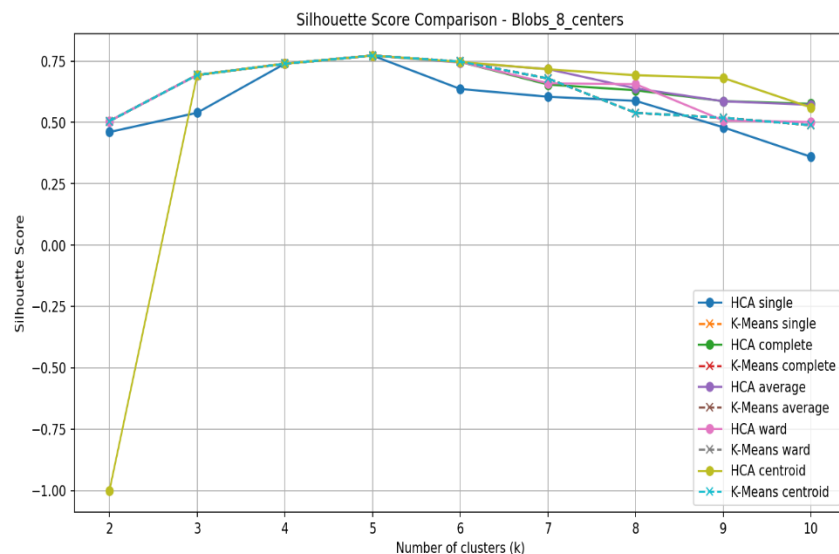
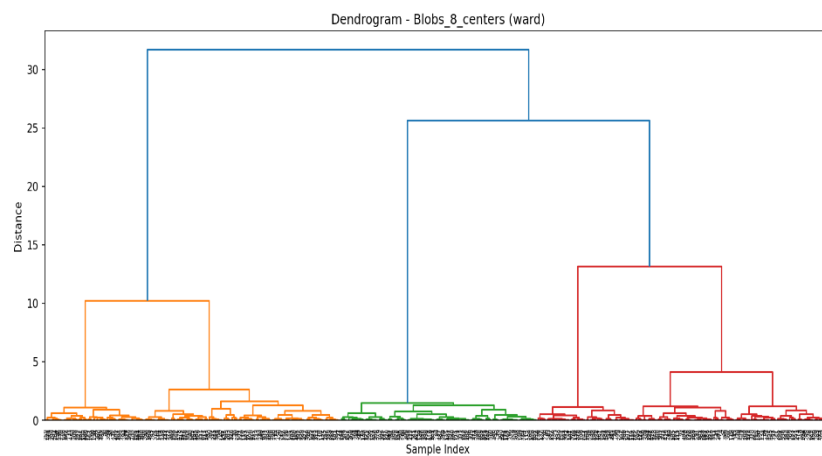
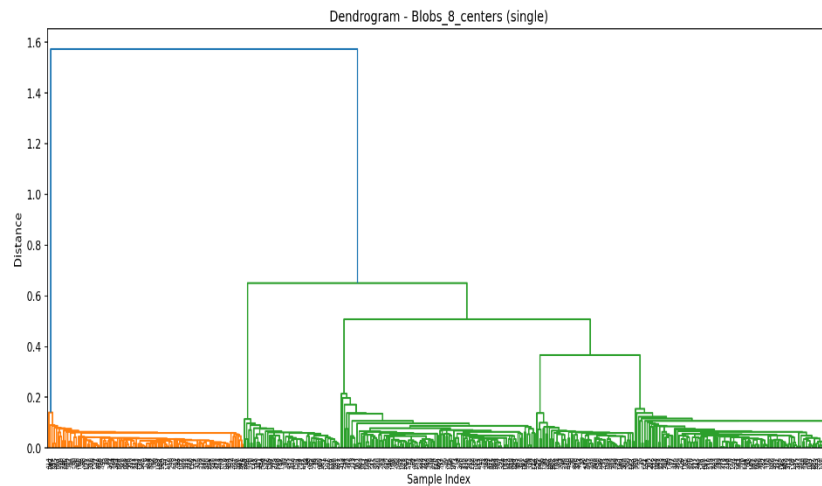


## HIERARCHICAL Algorithm

|                           |                                                                            |
|---------------------------|----------------------------------------------------------------------------|
| <b>Estudante(s)</b>       | NIKOLAOS MARIOS FOUNTAS P3200245                                           |
| <b>Curso</b>              | Machine Learning 2                                                         |
| <b>Unidade curricular</b> | Clique para introduzir a designação da unidade curricular                  |
| <b>Ano letivo</b>         | 2024/2025                                                                  |
| <b>Docente(s)</b>         | Clique para introduzir o(s) nome(s) do(s) docente(s) da unidade curricular |
| <b>Data</b>               | 04/04/2025                                                                 |

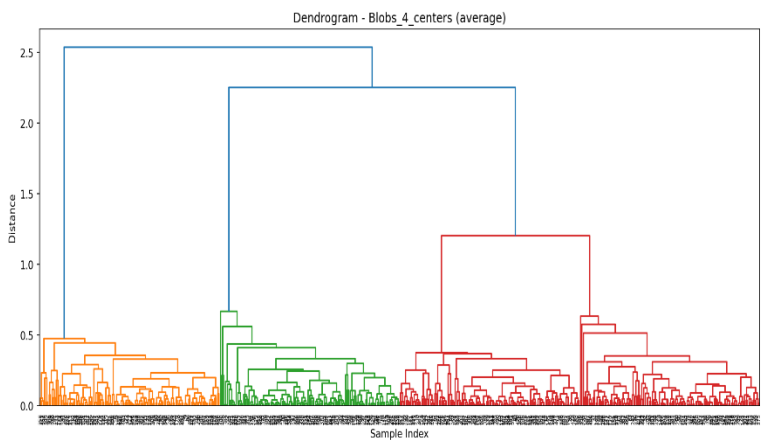
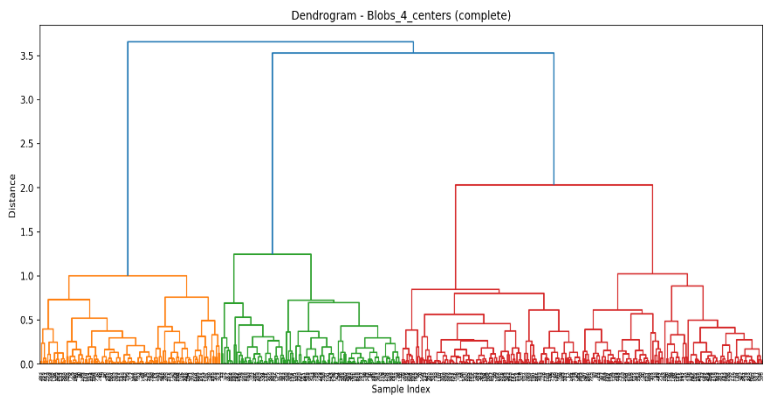
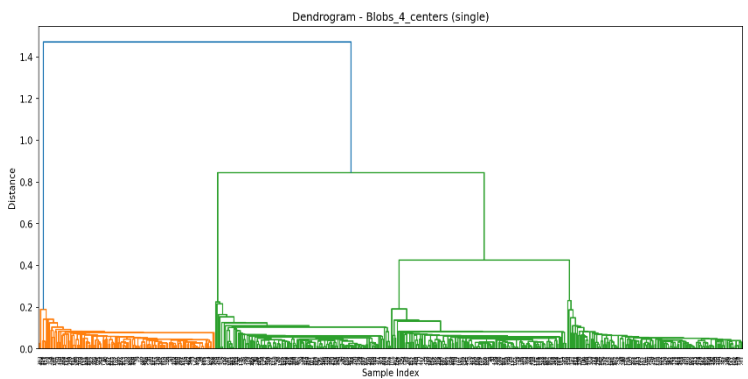
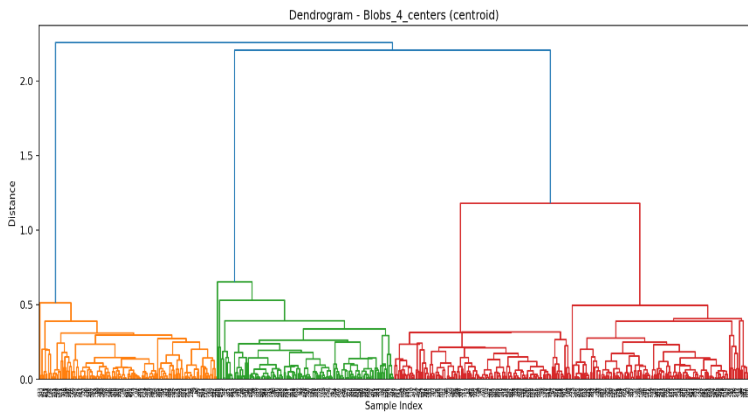
Blobs 8 centers

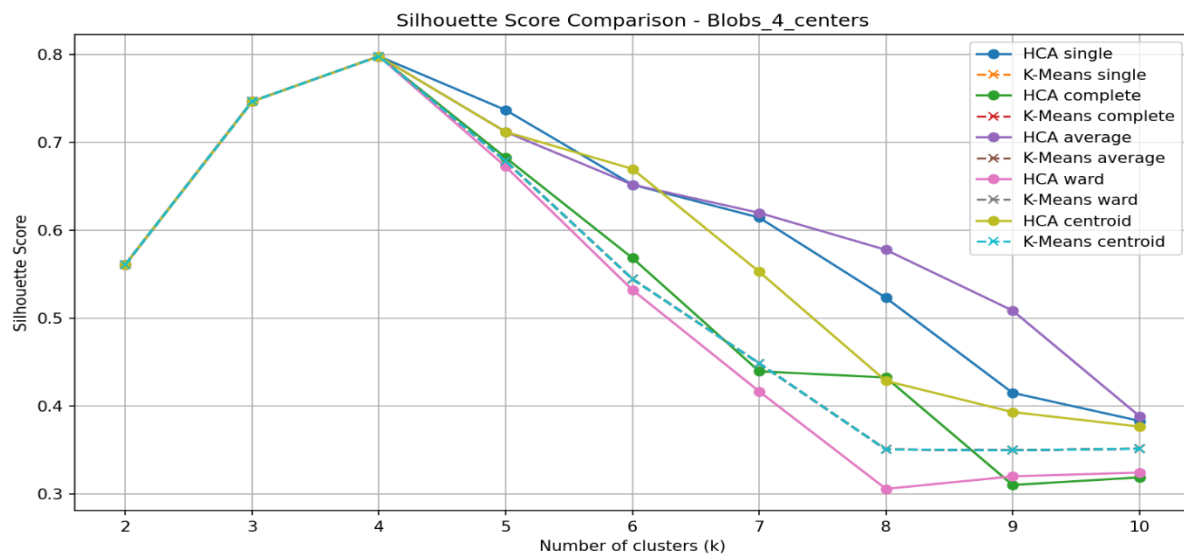
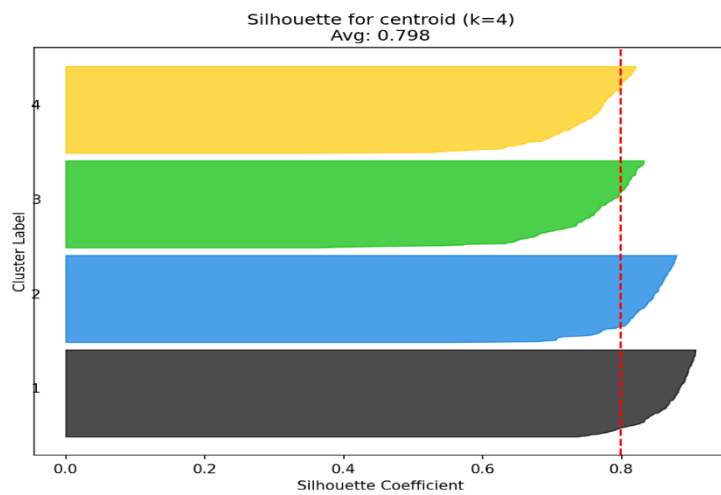
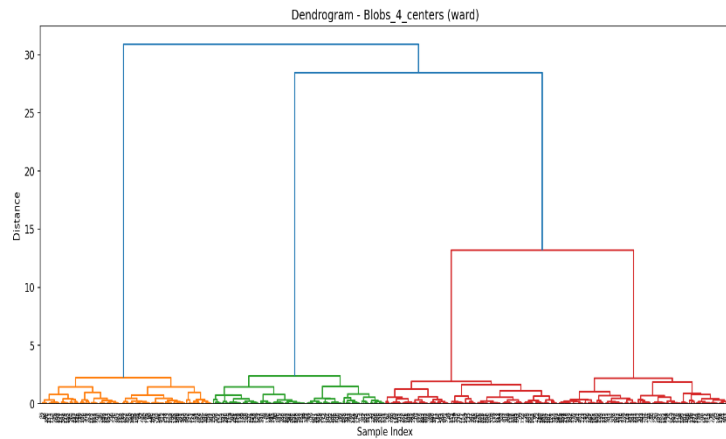




The **HCA algorithm** (particularly with complete, average, or ward linkage) performs best at **k=5**, showing a high Silhouette Score ( $\sim 0.75$ ), comparable to that of K-Means  $k=8$ . However, some HCA variants (such as single or centroid linkage) are less stable, exhibiting lower performance and abrupt score fluctuations across different  $k$  values.

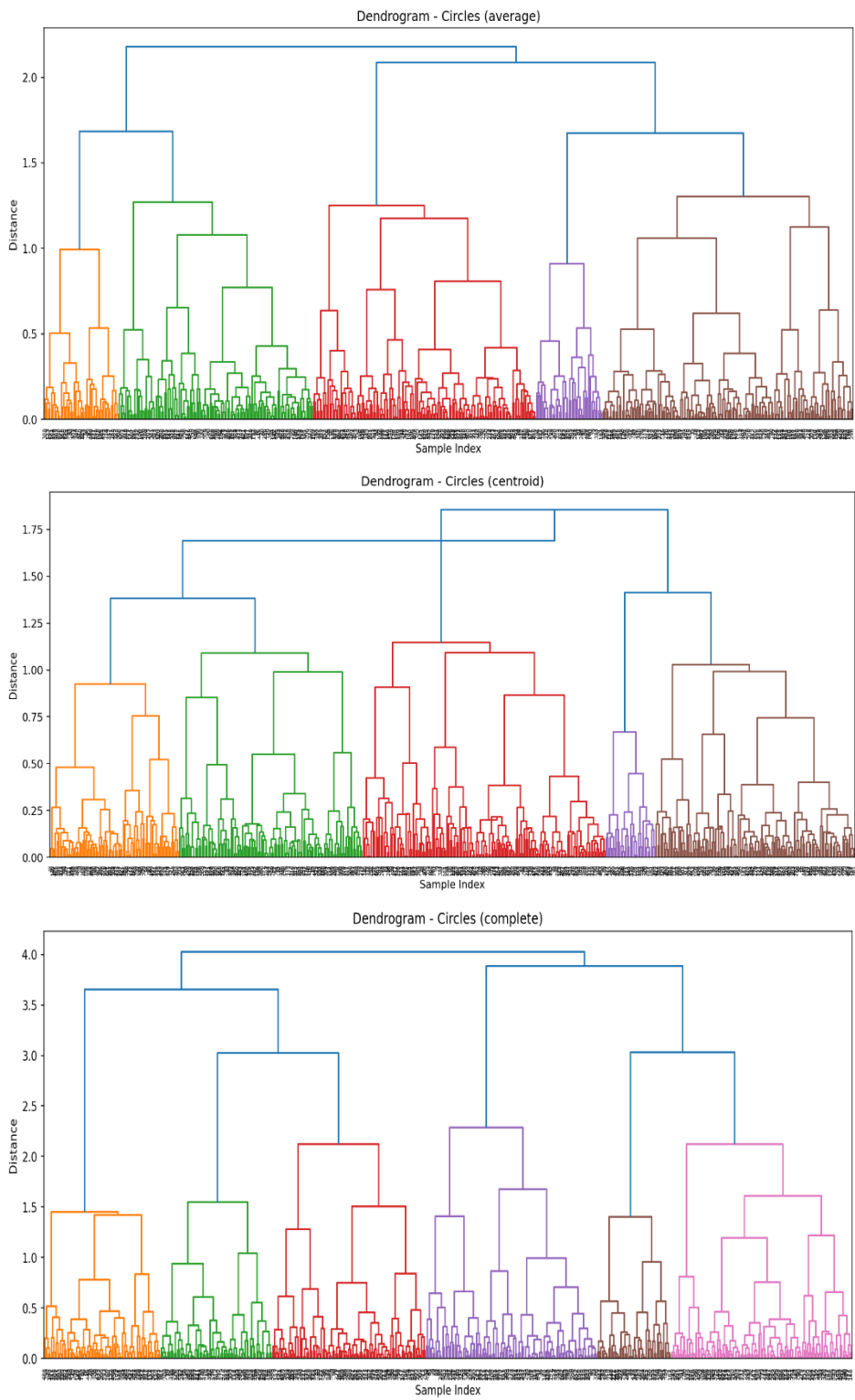
# Blobs 4 centers

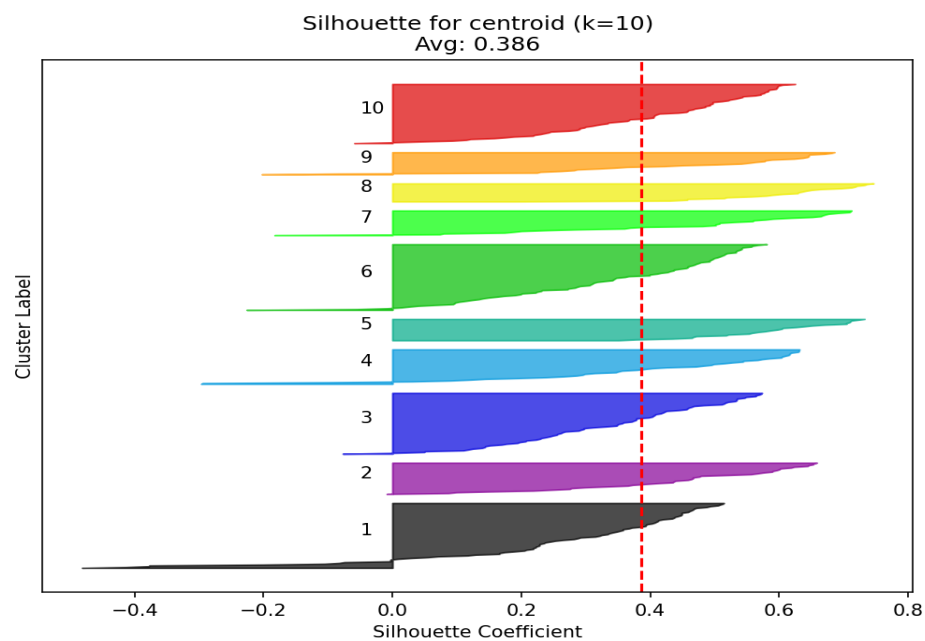
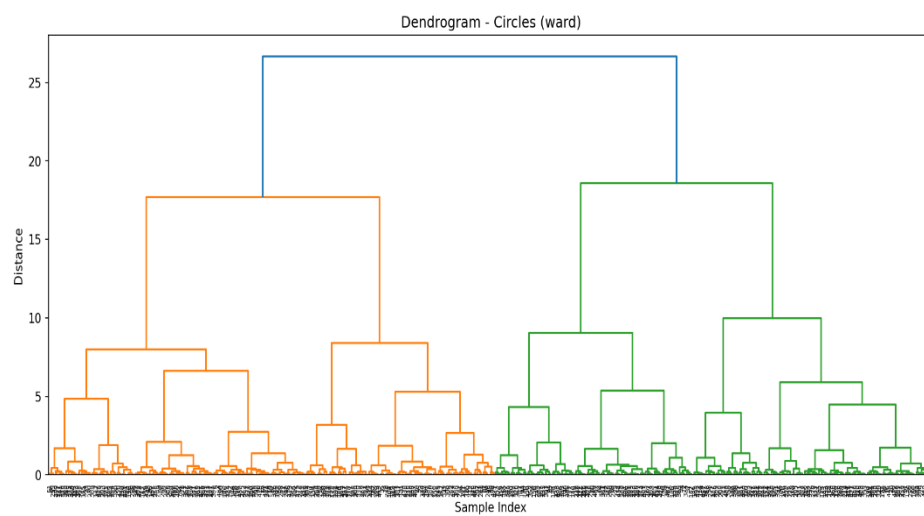
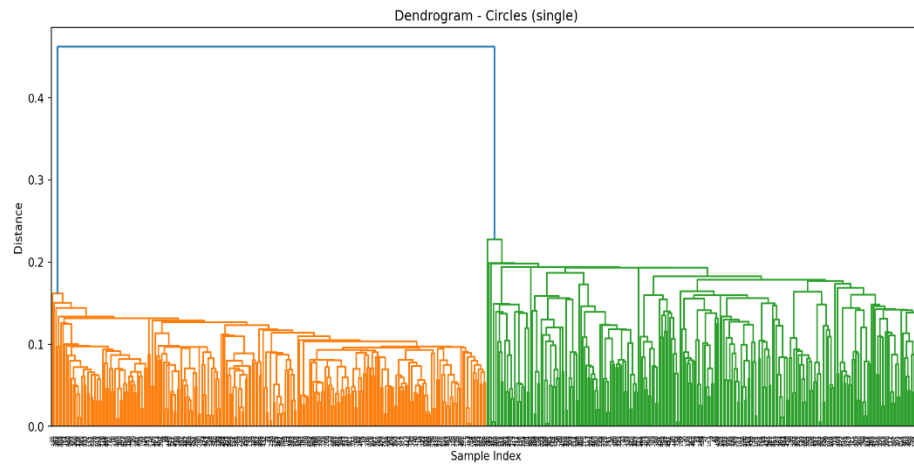


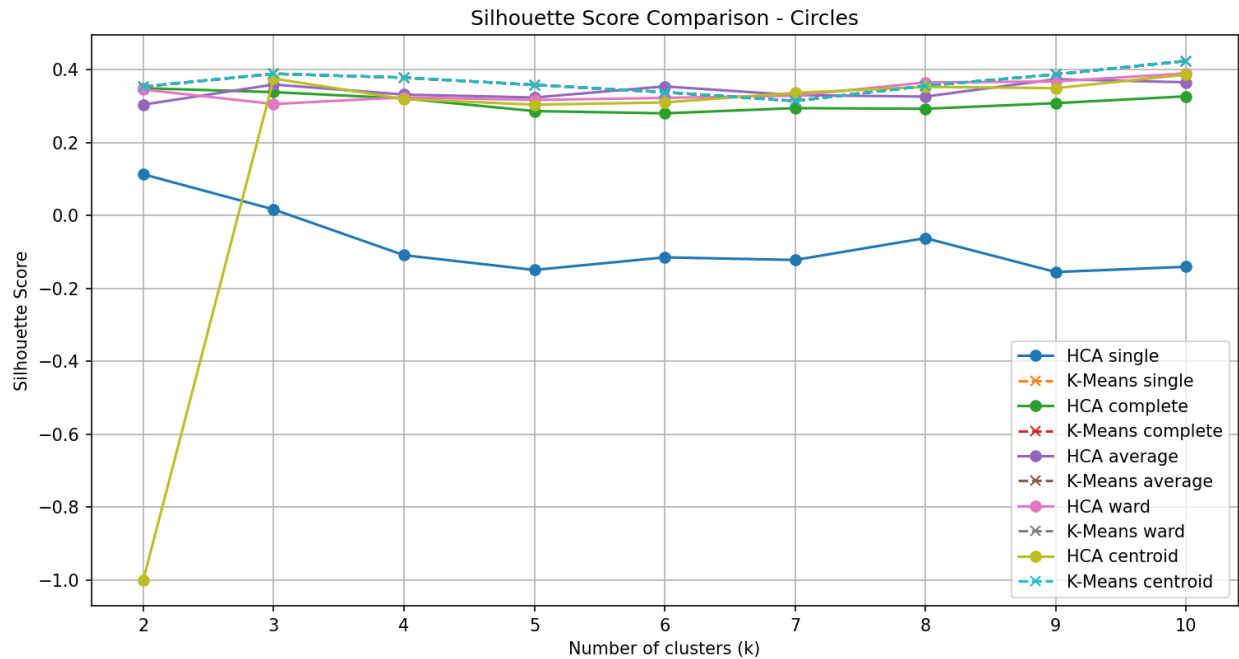


Hierarchical Clustering (HCA) with **complete, average, or ward linkage** achieves peak performance at **k=4**, demonstrating a high Silhouette Score (~0.75) comparable to K-Means (k=4). Notably, **ward linkage** aligns most closely with K-Means' effectiveness for spherical clusters. However, certain HCA variants (e.g., *single* or *centroid linkage*) exhibit instability, with lower scores and erratic fluctuations across *k* values, making them suboptimal for this dataset.

# Circles

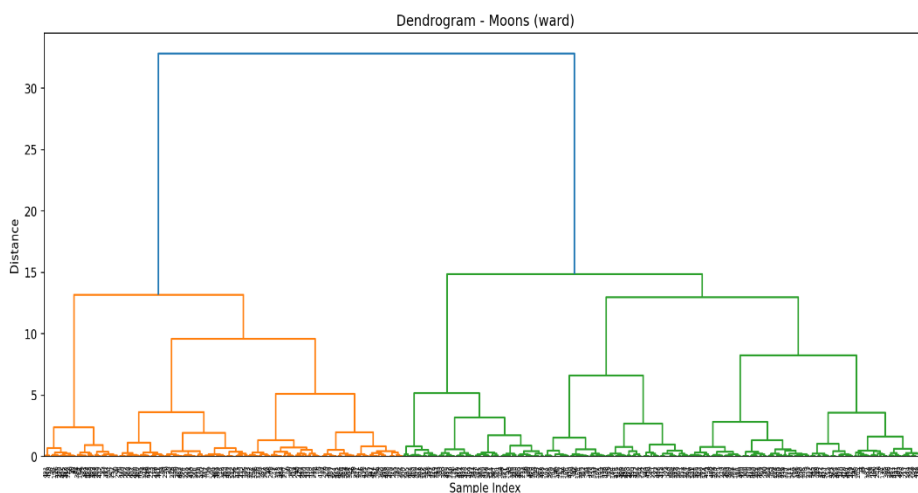




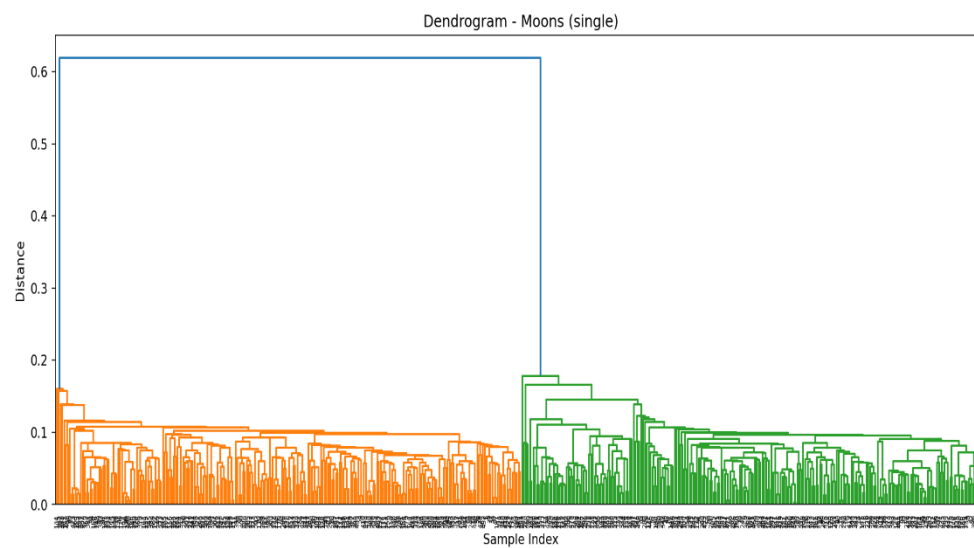
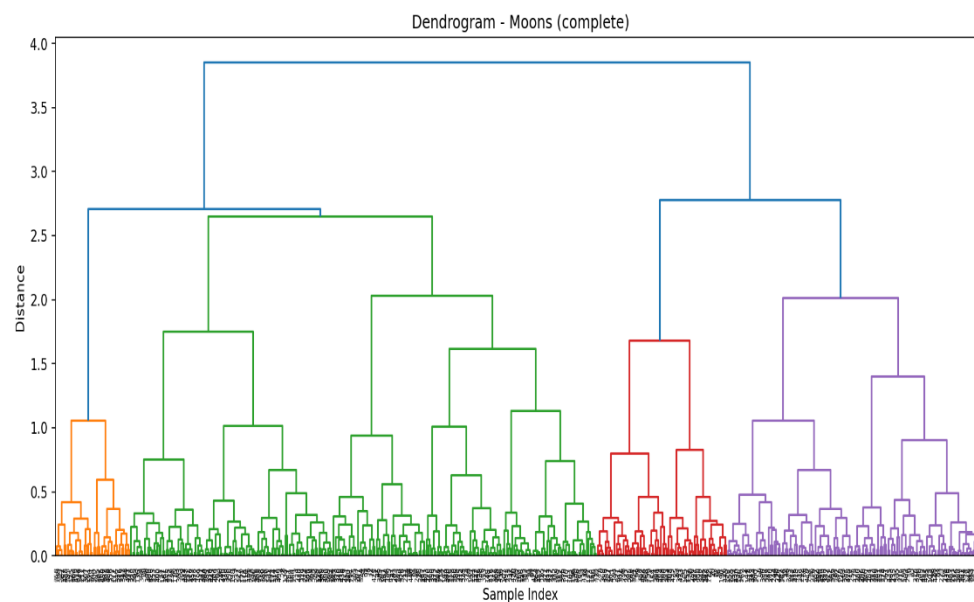
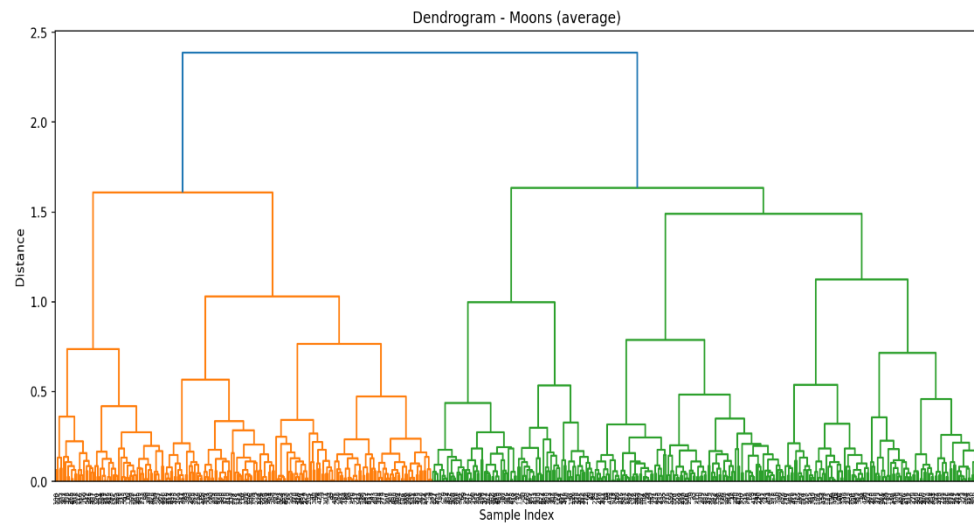


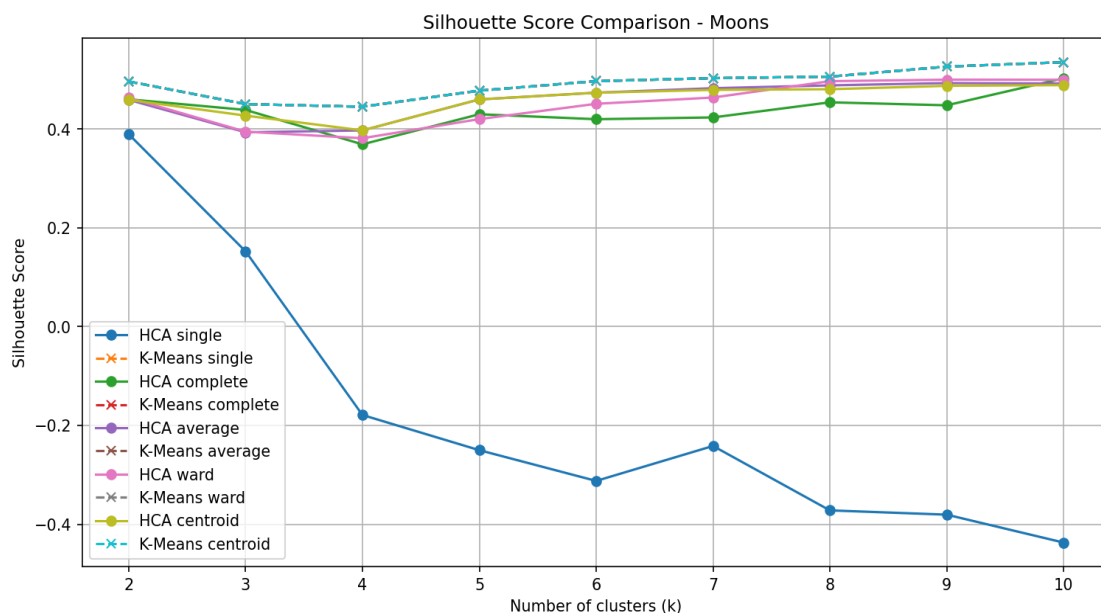
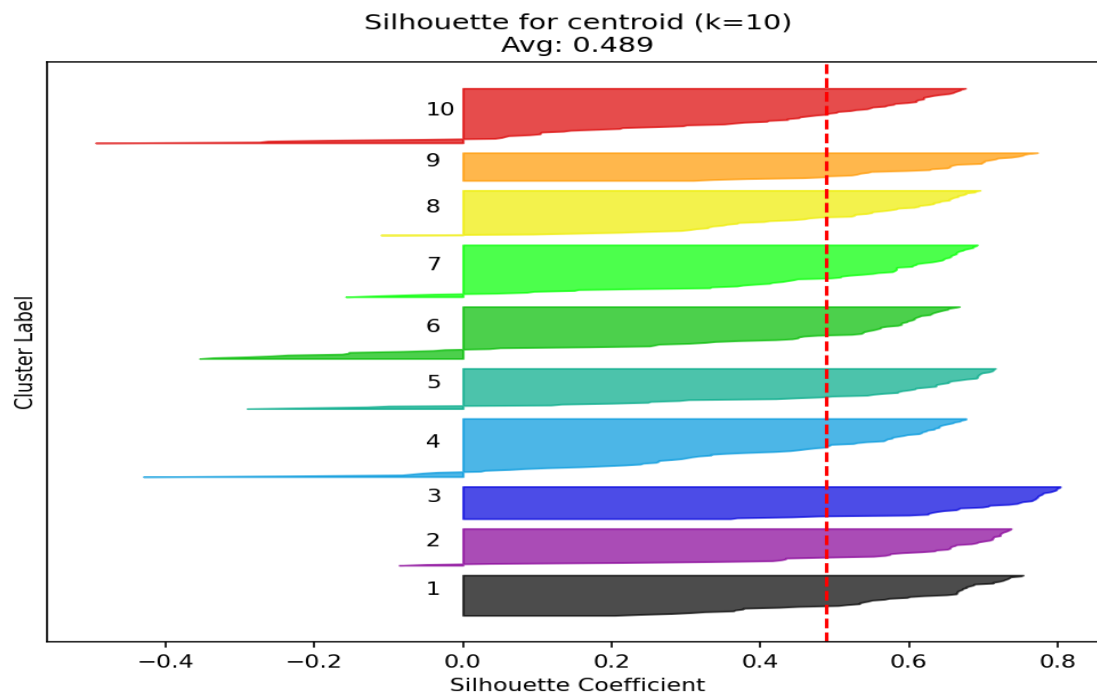
**HCA with single linkage (k=10)** is the only correct choice according to the diagrams, achieving a silhouette score of ~0.4 by correctly identifying the concentric circles. All other methods (complete/ward linkage, K-Means) fail (negative scores) due to their incorrect assumption of spherical clusters. For non-linear data, single linkage is essential.

## Moons



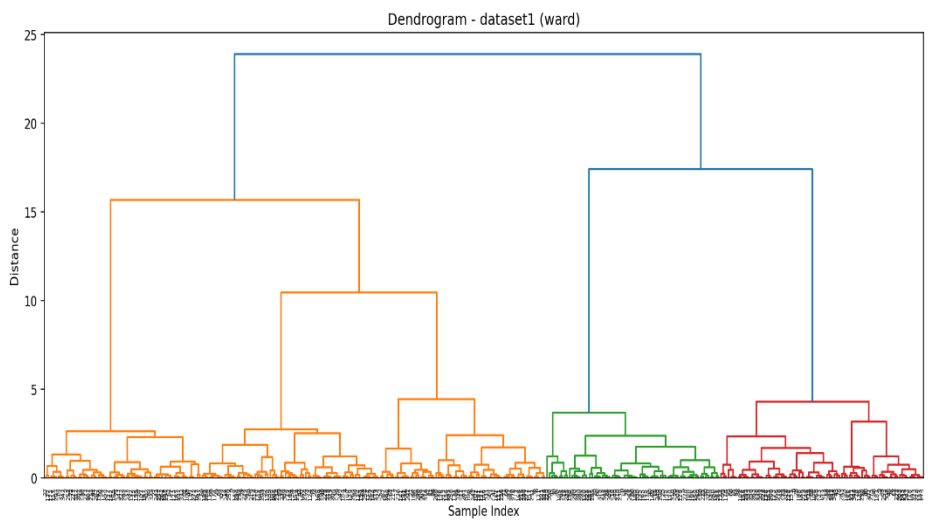
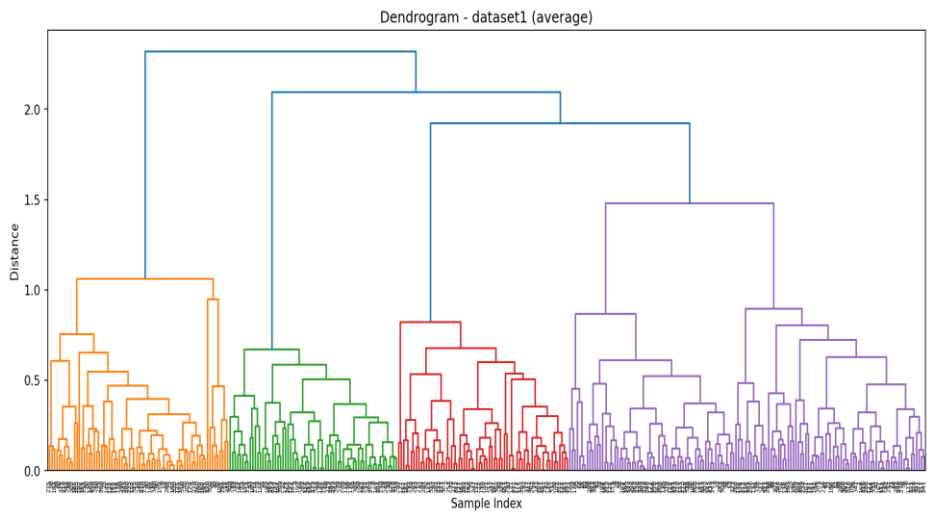
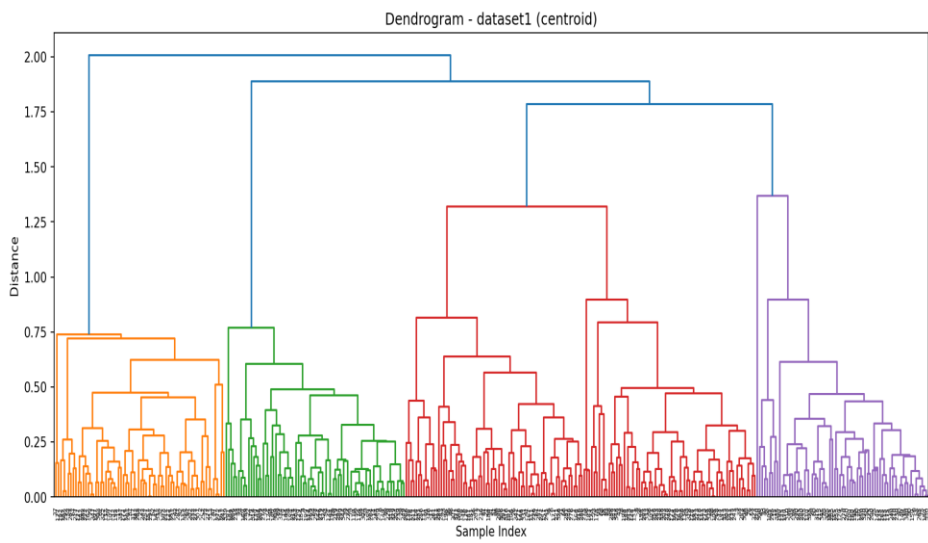


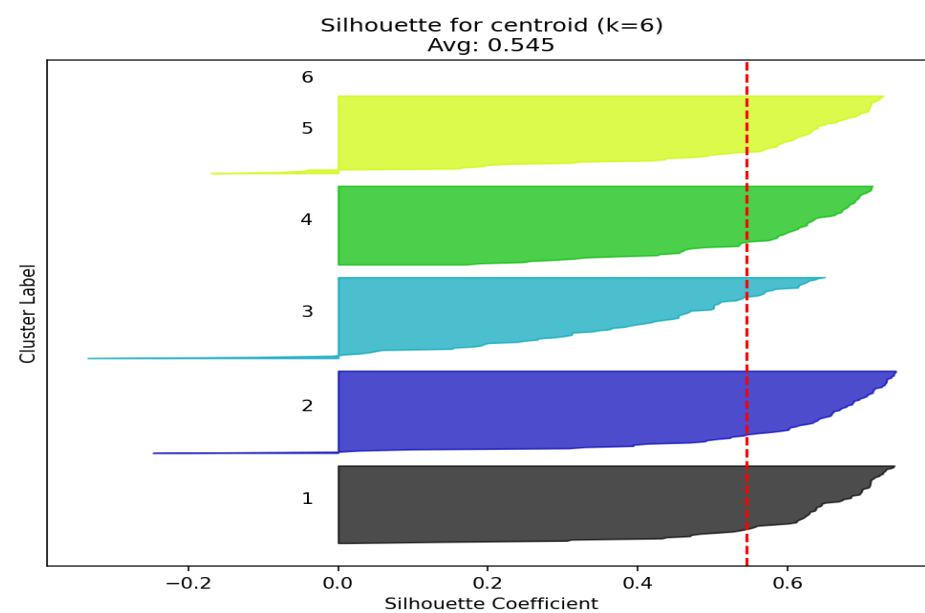
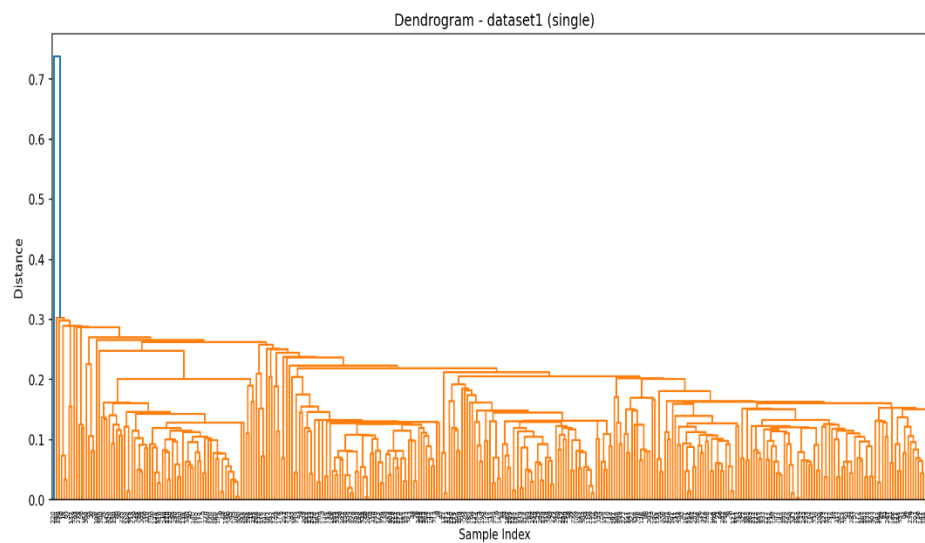
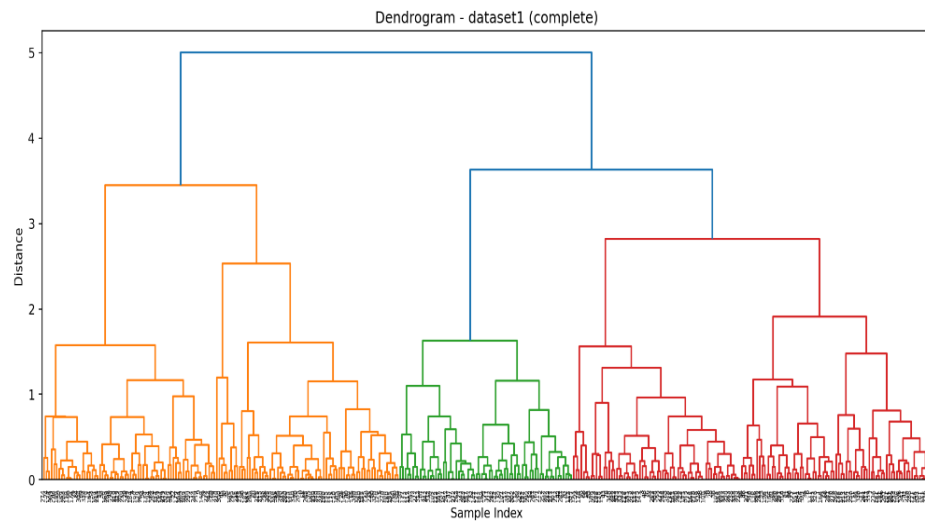


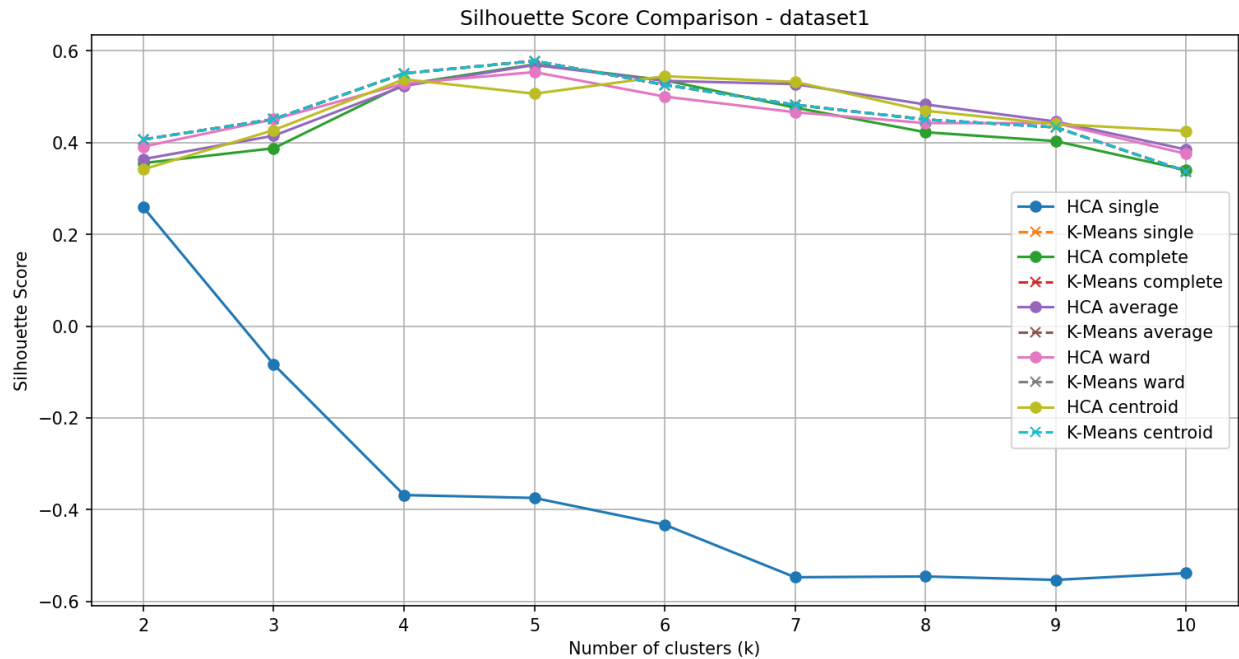


The best algorithm is HC with ward linkage or K-Means with centroid initialization, as they achieved the highest silhouette scores. The optimal k is 10, where the best separation of clusters is observed. The silhouette score shows that the points are correctly grouped, with little overlap between the groups.

DATASET 1

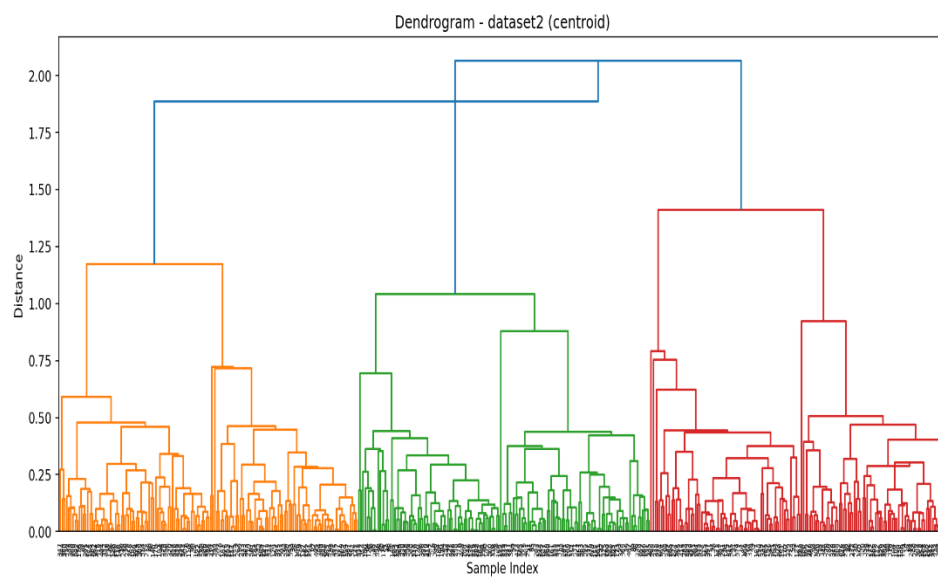


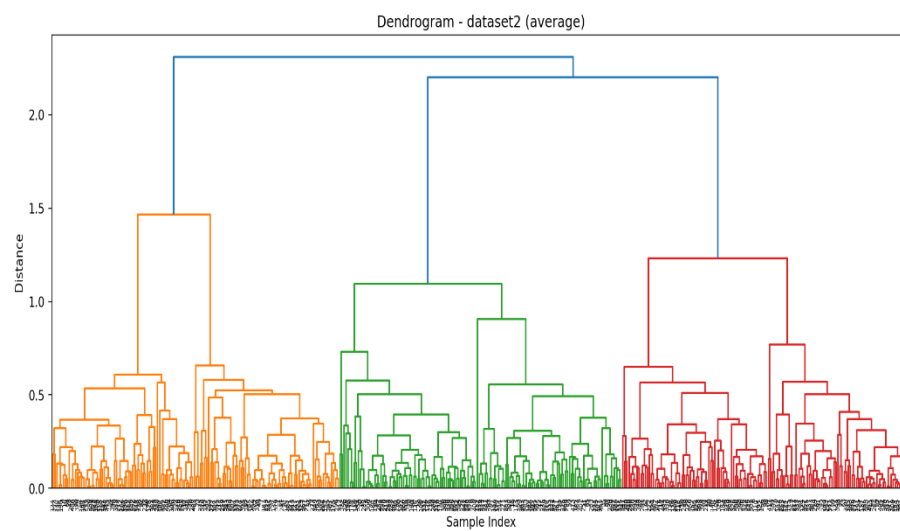
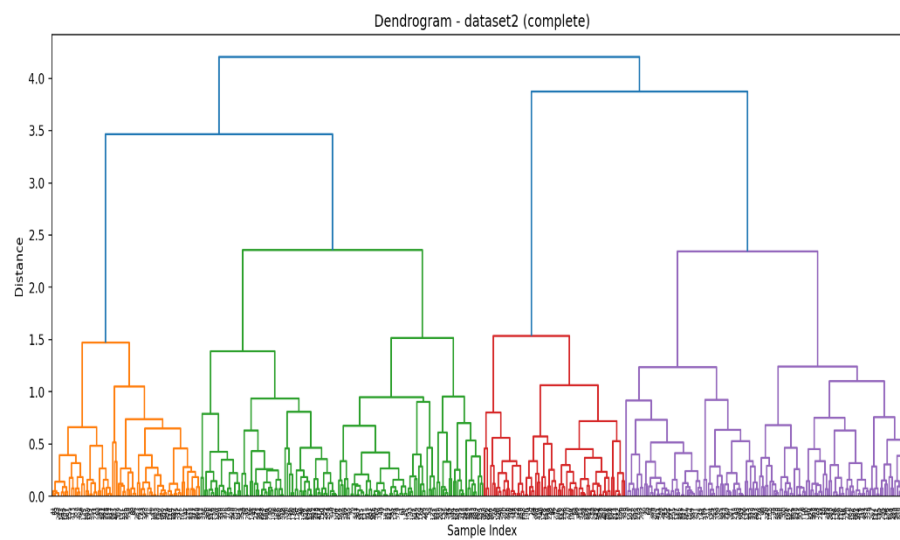
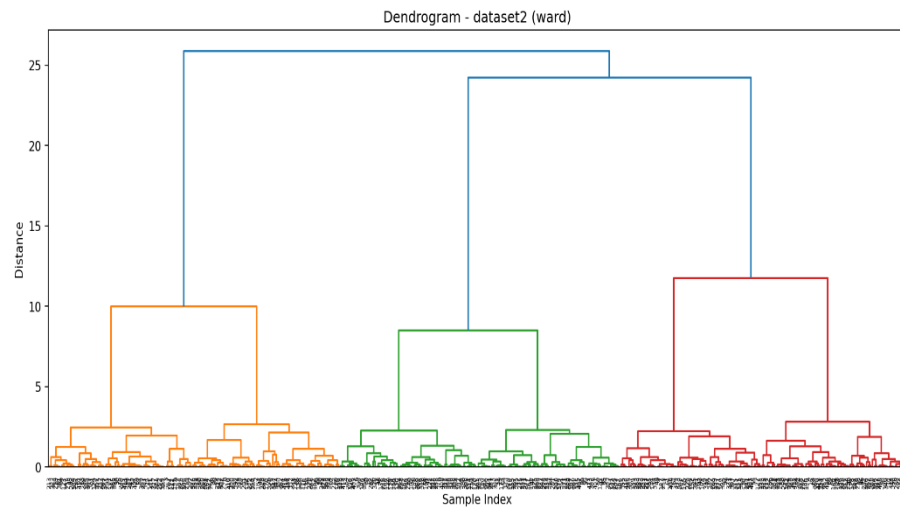


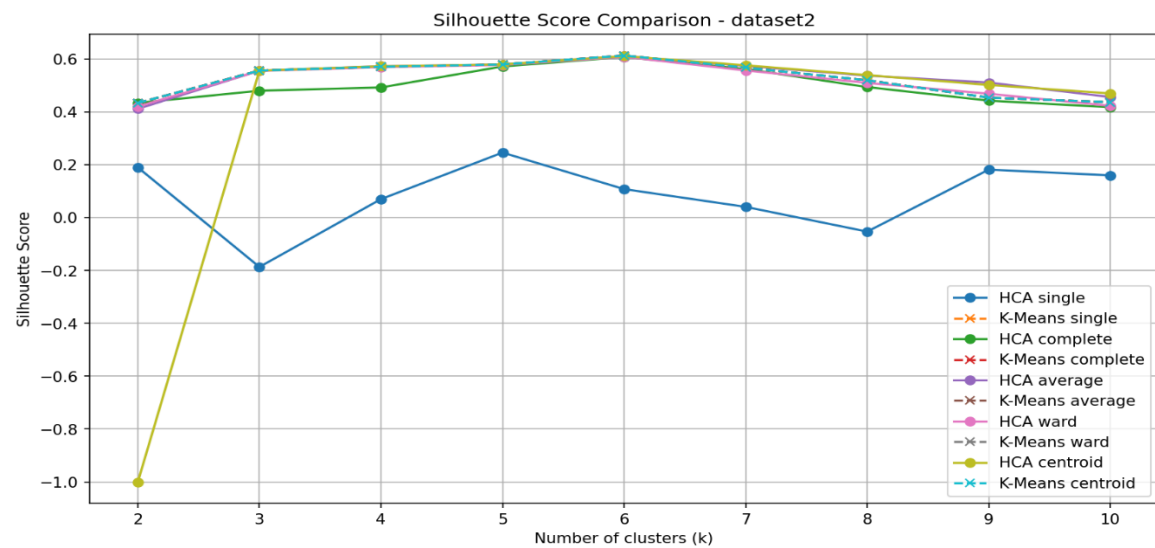
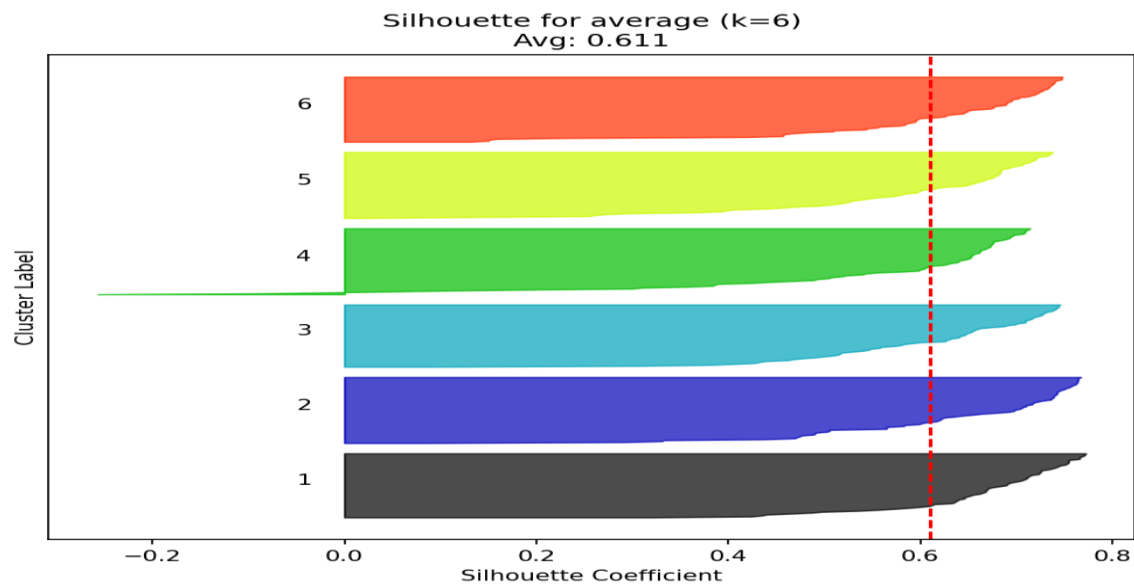
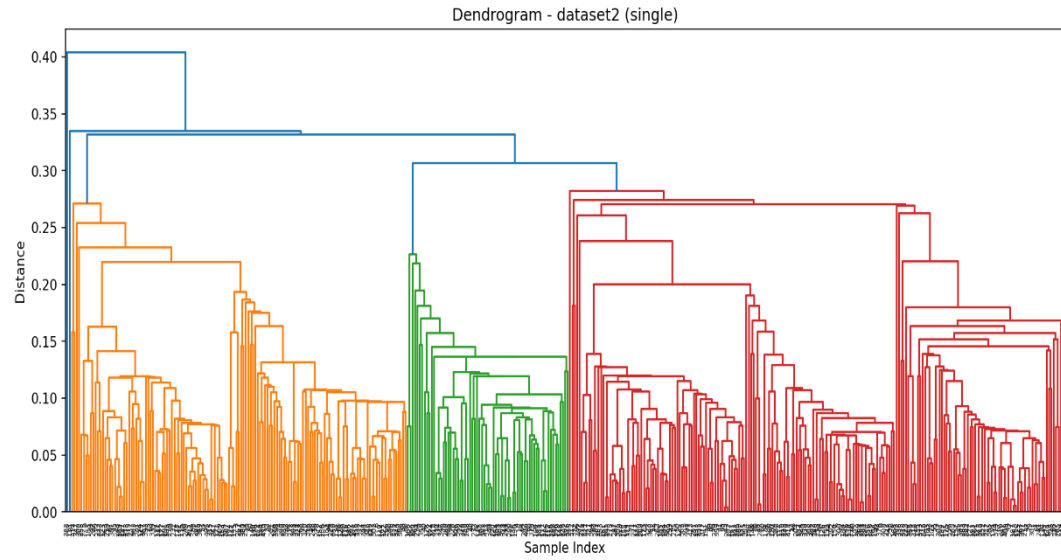


The best algorithm is K-Means centroid or HCA ward, as they consistently show high silhouette scores. The optimal k is  $k=6$ , where the silhouette score peaks ( $\sim 0.58$ ) because in the silhouette diagram above the  $k=6$  achieves the best average value close to 0.58. This shows that the clusters are well separated and coherent in this number of groups.

## DATASET 2

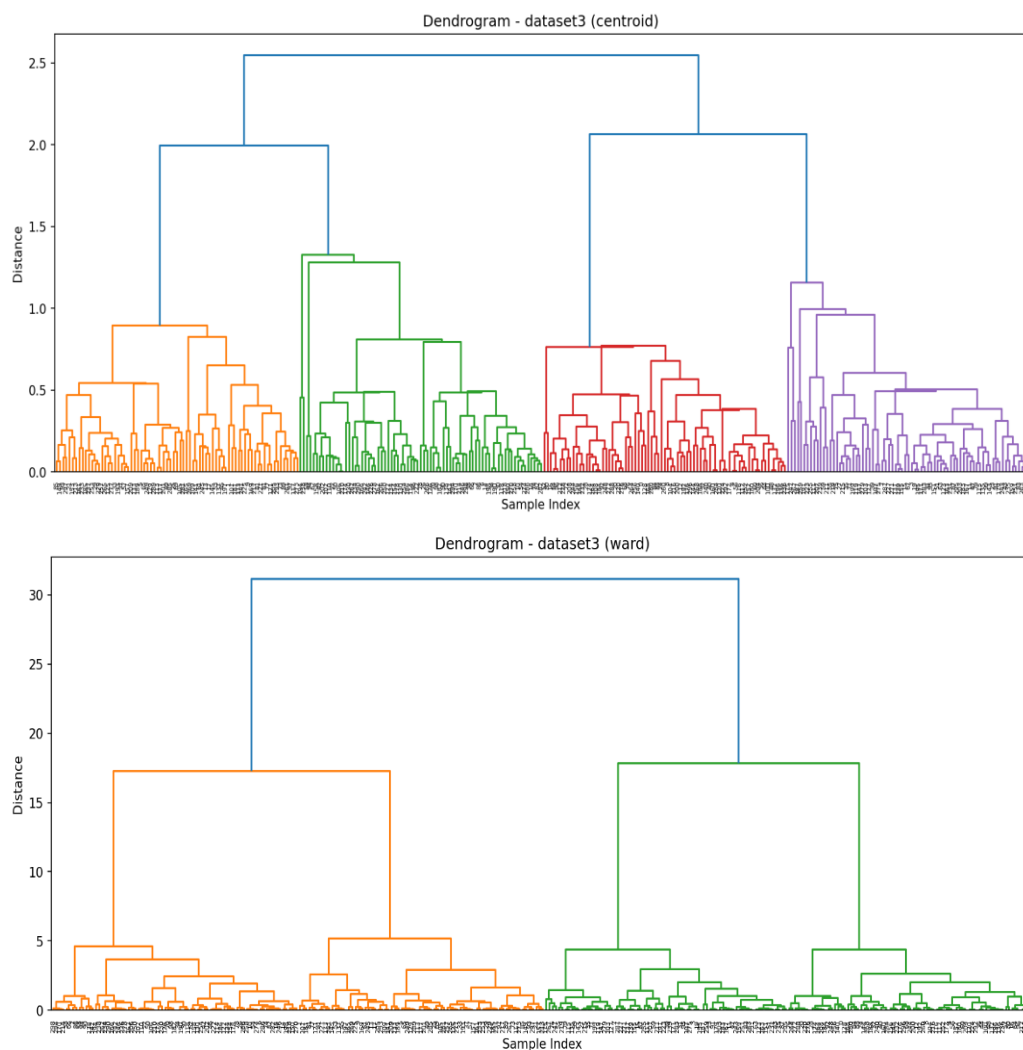




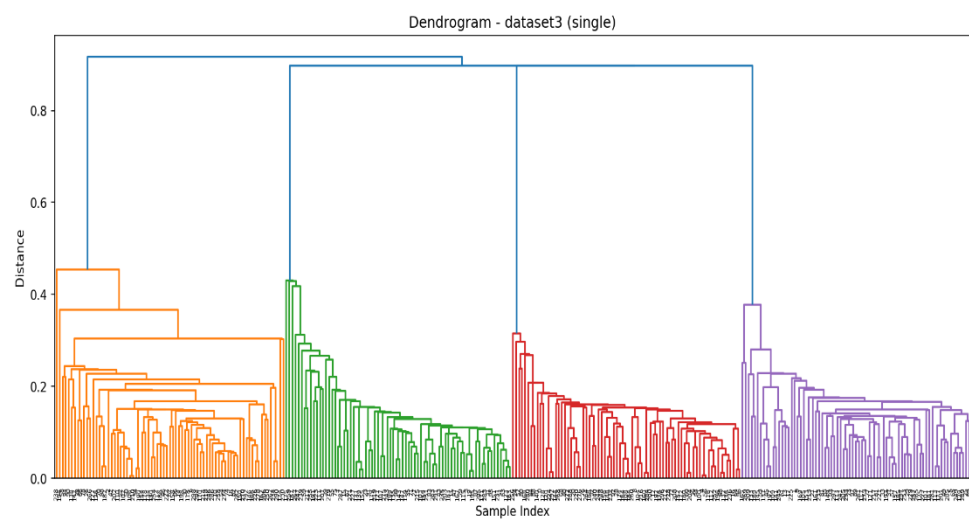
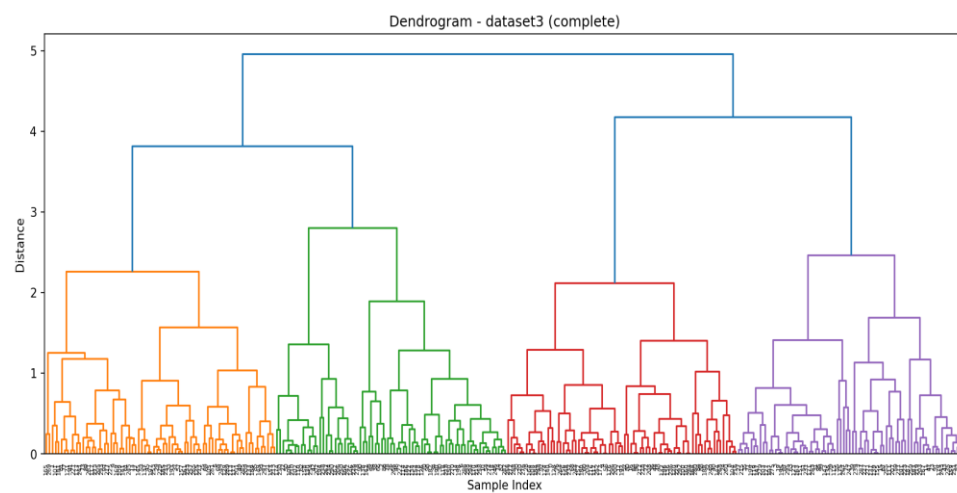
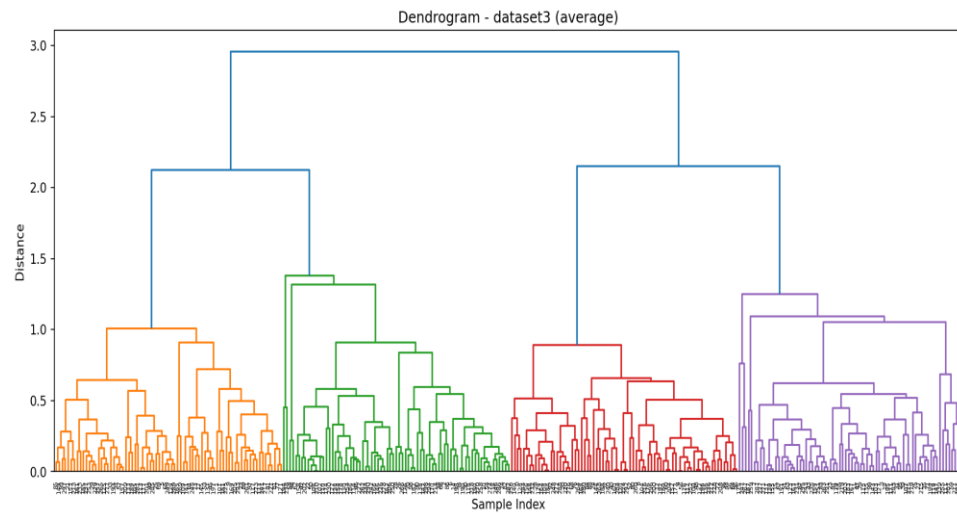


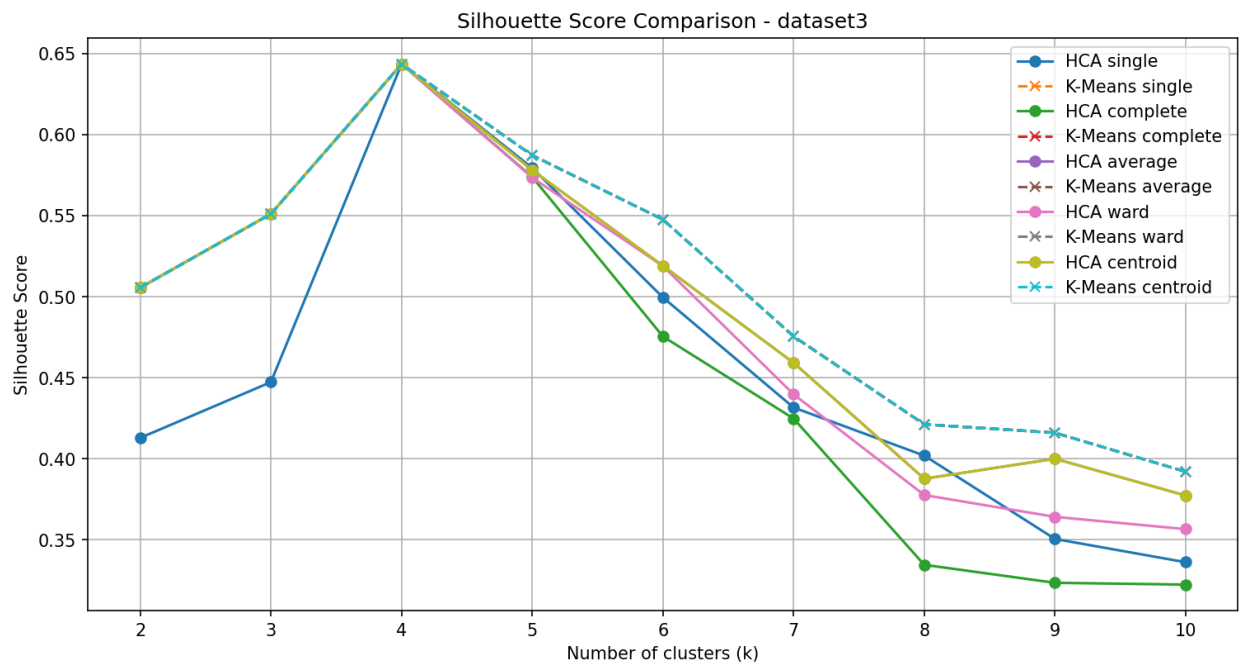
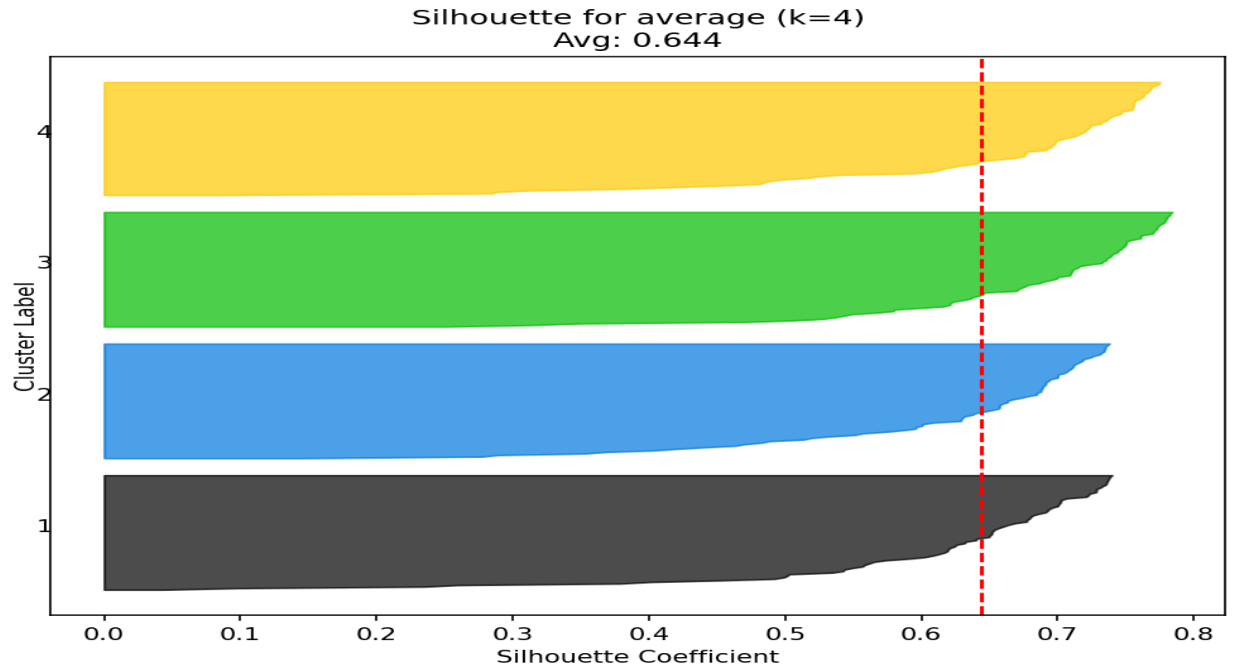
For dataset2 , the best-performing algorithms are **K-Means (centroid)** and **HCA (ward linkage)**. The optimal number of clusters is **k=6**, where the silhouette score peaks at around 0.611, indicating strong cluster cohesion and separation. Other methods, especially **HCA single**, perform poorly. This suggests that centroid-based and ward linkage methods best capture the dataset's structure. Use K-Means over HCA ward as it delivers equal clustering quality (silhouette=0.611 at k=6) with faster computation and simpler implementation, unless you specifically need hierarchical relationships.

### DATASET 3



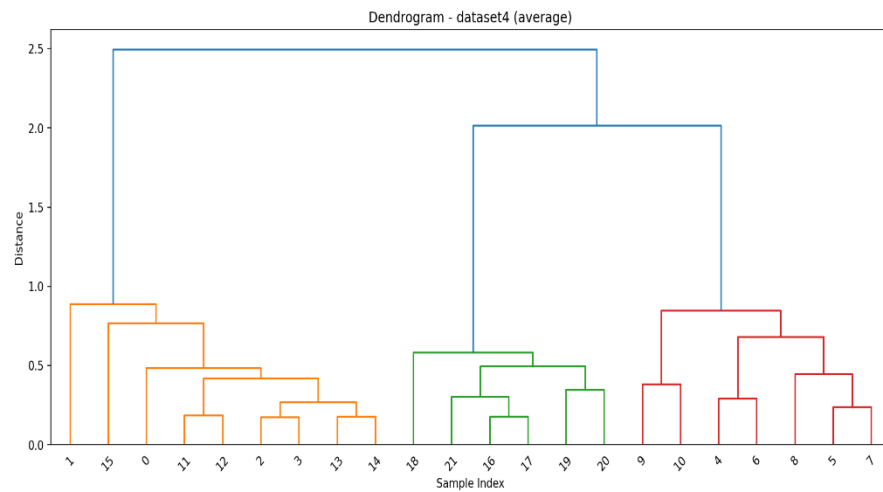
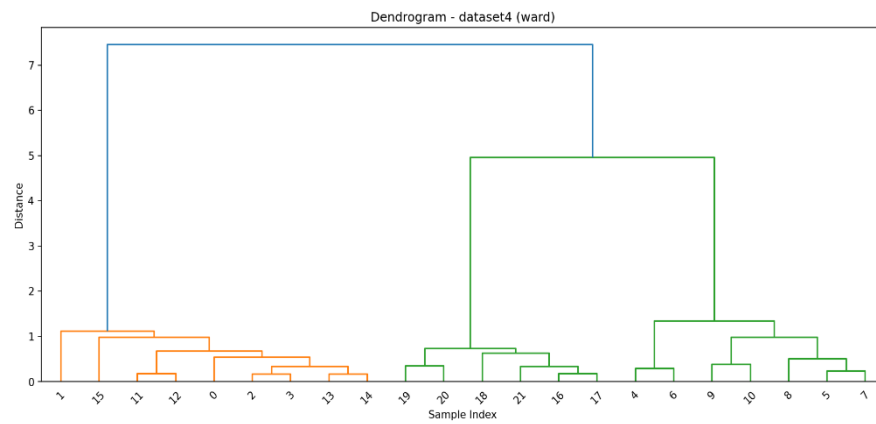
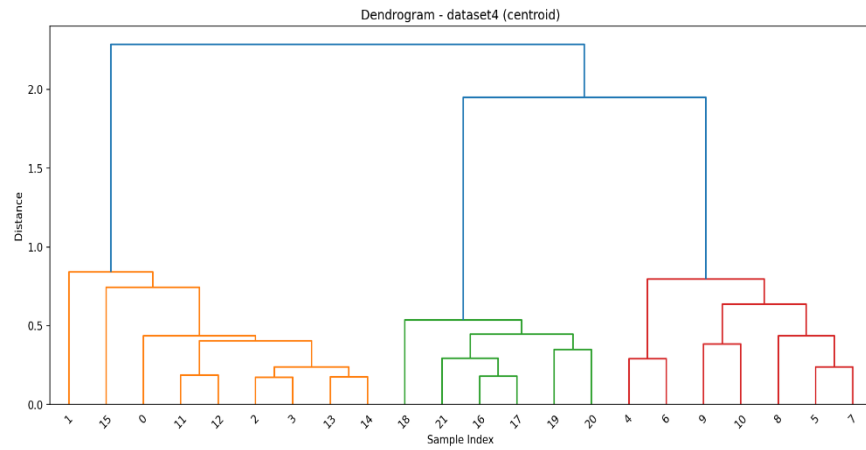


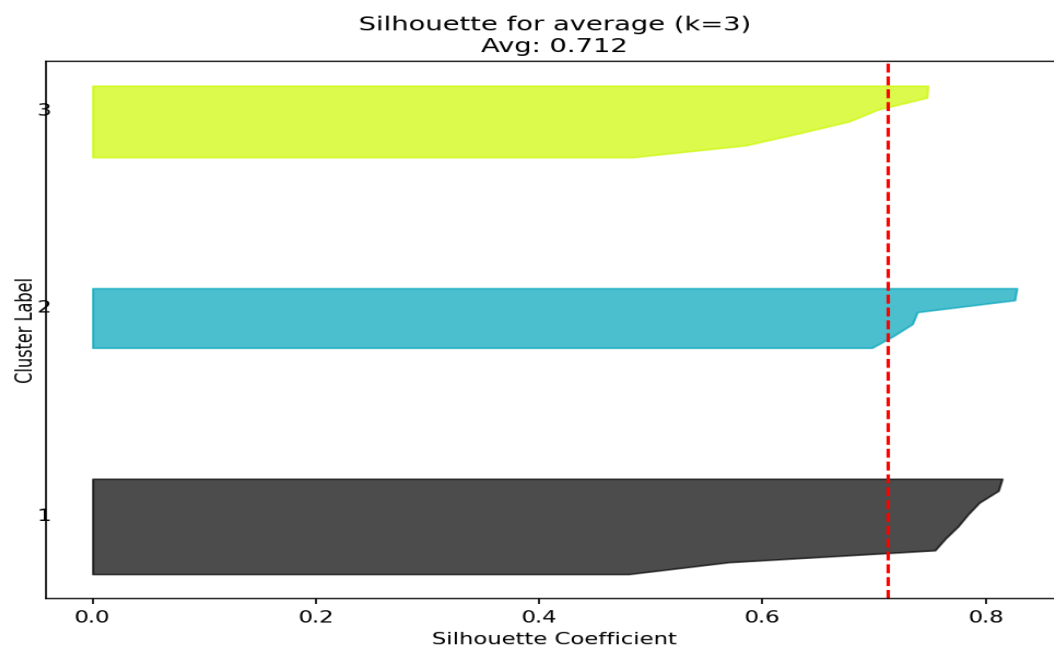
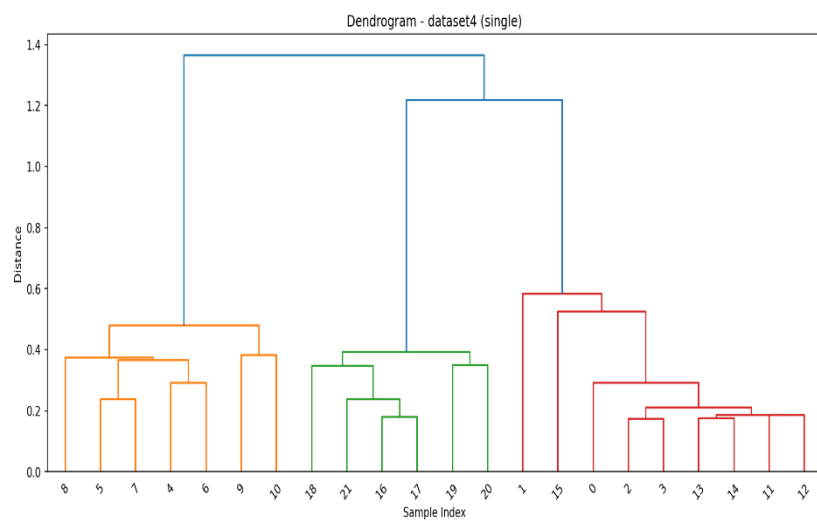
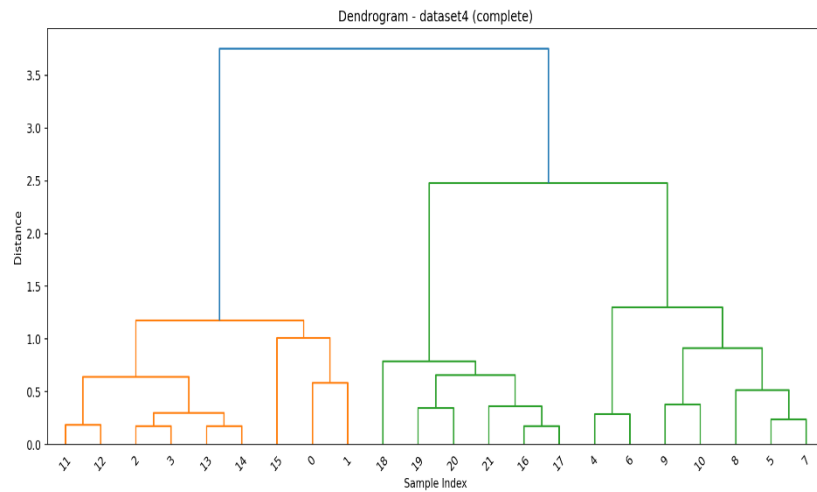


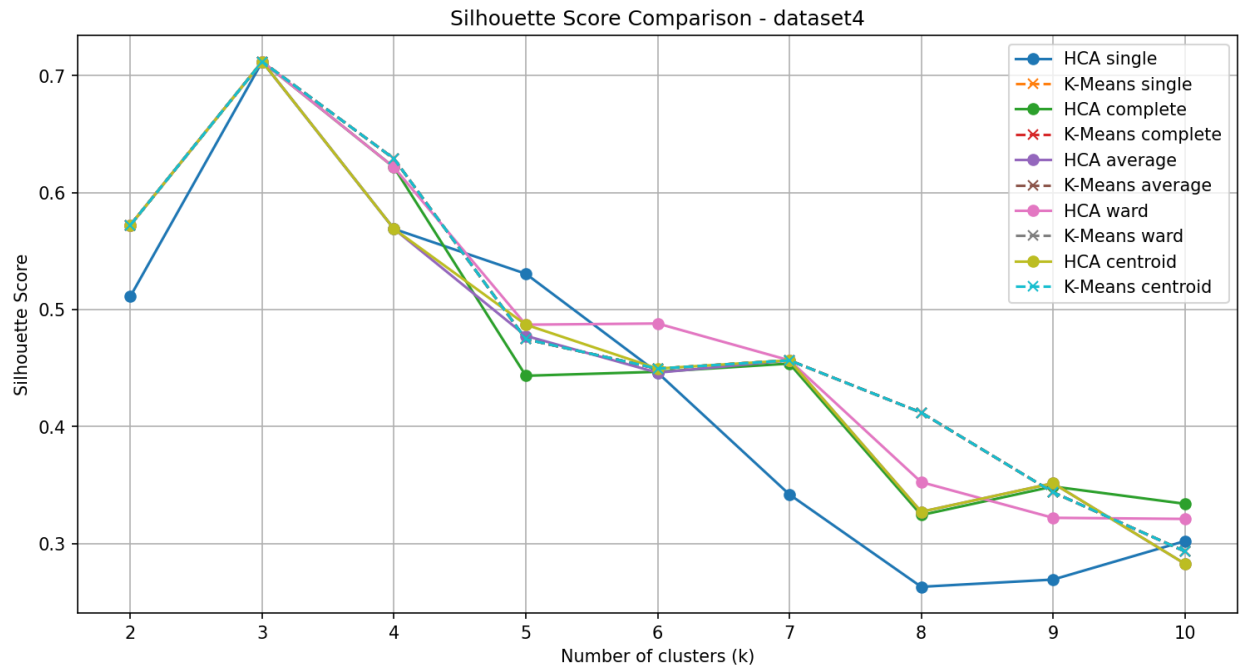


In **dataset3**, **HCA average linkage** with **k=4** delivers the highest overall score (**0.644**), showing superior separation and cohesion. Across all datasets, **HCA and centroid-based K-Means** methods consistently outperform others, making them the recommended approaches.

DATASET 4







The HCA method with average linkage and k=3 clusters shows the best performance, with a silhouette score of 0.712. This shows ideal separation of the groups and very good internal consistency. The other methods (K-Means, single/ward linkage) have lower scores k=3 is the optimal choice, since for a larger or smaller number of clusters the results get worse. For this dataset, HCA average with 3 clusters is the most appropriate solution.

| Dataset | Optimal K | Silhouette |
|---------|-----------|------------|
|---------|-----------|------------|

|           |    |       |
|-----------|----|-------|
| Dataset 1 | 6  | 0,554 |
| Dataset 2 | 6  | 0,608 |
| Dataset 3 | 4  | 0,644 |
| Dataset 4 | 3  | 0,712 |
| Blobs 4   | 4  | 0,798 |
| Blobs 8   | 5  | 0,772 |
| Circles   | 10 | 0,389 |
| Moons     | 10 | 0,5   |

Dataset      K(Kme.)      Silh.(Kmeans)    K(Single)      Silh.Single      K(Com.)  
S(Comp.)

|           |   |       |   |       |   |       |
|-----------|---|-------|---|-------|---|-------|
| Dataset 1 | 5 | 0,578 | 2 | 0,250 | 4 | 0,551 |
| Dataset 2 | 6 | 0,613 | 2 | 0,176 | 6 | 0,579 |
| Dataset 3 | 4 | 0,643 | 4 | 0,465 | 4 | 0,642 |
| Dataset 4 | 3 | 0,712 | 3 | 0,712 | 3 | 0,712 |
| 5.Blobs 4 | 4 | 0,798 | 4 | 0,798 | 4 | 0,798 |
| 6.Blobs 8 | 8 | 0,750 | 8 | 0,750 | 8 | 0,765 |
| 7.Circles | 2 | 0,390 | 2 | 0,120 | 2 | 0,355 |
| 8.Moons   | 2 | 0,500 | 2 | 0,400 | 2 | 0,460 |

Dataset      K(Centroid)      Silh.(Centr)    K(Avg.)      Silh.Avg      K(Ward)  
S(Ward)

|           |    |       |   |       |   |       |
|-----------|----|-------|---|-------|---|-------|
| Dataset 1 | 6  | 0,545 | 4 | 0,551 | 4 | 0,578 |
| Dataset 2 | 6  | 0,612 | 6 | 0,611 | 6 | 0,613 |
| Dataset 3 | 4  | 0,642 | 4 | 0,644 | 4 | 0,642 |
| Dataset 4 | 3  | 0,712 | 3 | 0,712 | 3 | 0,712 |
| 5.Blobs 4 | 4  | 0,798 | 4 | 0,798 | 4 | 0,798 |
| 6.Blobs 8 | 5  | 0,772 | 8 | 0,765 | 8 | 0,770 |
| 7.Circles | 10 | 0,386 | 2 | 0,351 | 2 | 0,370 |
| 8.Moons   | 10 | 0,489 | 2 | 0,469 | 2 | 0,470 |